

Self-Adjoint (Sturm–Liouville) Form for Second-Order ODEs

Goal. Record the basic facts showing that any linear second-order scalar ODE on an interval reduces to self-adjoint (Sturm–Liouville) form, and deduce standard spectral consequences for regular boundary value problems.

Note. Feel free to ignore Section 3. It is for recording some useful facts which we shall prove later in class.

1. Reduction to Sturm–Liouville form

Consider

$$y'' + P(t)y' + Q(t)y = \lambda W(t)y, \quad t \in (a, b), \quad (1)$$

with continuous P, Q, W and $W > 0$. Define the integrating factor

$$p(t) := \exp\left(\int_{t_0}^t P(s) ds\right) \quad (> 0).$$

Then

$$(p y')' = p y'' + p P y' = p (y'' + P y').$$

Multiplying (1) by p gives

$$(p y')' + p Q y = \lambda p W y,$$

i.e.

$$-(p y')' + q y = \lambda w y, \quad q := p Q, \quad w := p W. \quad (2)$$

Thus every such ODE can be put into Sturm–Liouville form with coefficients (p, q, w) and $p > 0$, $w > 0$.

2. Lagrange (Green) identity and adjoint

Let $L[y] := -(p y')' + q y$ with p, q as above. For $u, v \in C^1[a, b]$ with $p u', p v' \in C^1[a, b]$,

$$\int_a^b (v L[u] - u L[v]) dt = [p(v u' - u v')]_{t=a}^{t=b}. \quad (3)$$

In particular, with the weighted inner product

$$\langle u, v \rangle_w := \int_a^b u \bar{v} w dt,$$

and boundary conditions that kill the endpoint term in (3), L is formally self-adjoint. The standard *separated* self-adjoint boundary conditions at each endpoint $t = a$ or b are of the form

$$\alpha y + \beta p y' = 0 \quad (\alpha, \beta \in \mathbb{R} \text{ not both } 0). \quad (4)$$

3. Spectral consequences on a compact interval

Assume a *regular* problem: p, q, w continuous on $[a, b]$, $p > 0$, $w > 0$, with separated self-adjoint BCs (4) at a and b .

Theorem 1 (Reality and discreteness). *All eigenvalues λ of $L[y] = \lambda wy$ under (4) are real and form a discrete sequence $\lambda_1 < \lambda_2 < \dots \rightarrow +\infty$.*

Proof sketch. For $y \not\equiv 0$ satisfying the BCs,

$$\mathcal{R}[y] := \frac{\int_a^b (p|y'|^2 + q|y|^2) dt}{\int_a^b w|y|^2 dt}$$

is real. Minimization on the unit $L^2(w)$ -sphere and standard compactness (Rellich on $[a, b]$) yield a first eigenpair (λ_1, y_1) with $\lambda_1 = \min \mathcal{R}$. Orthogonalization in $L^2(w)$ and the min-max principle give an increasing, diverging sequence of real eigenvalues. $\square$

Theorem 2 (Orthogonality). *Eigenfunctions corresponding to distinct eigenvalues are orthogonal in $L^2([a, b], w dt)$: if $L[y_i] = \lambda_i wy_i$ with $\lambda_1 \neq \lambda_2$, then*

$$\int_a^b y_1 y_2 w dt = 0.$$

Proof. Apply (3) with $u = y_1$, $v = y_2$; the boundary term vanishes by the BCs, and $vL[u] - uL[v] = (\lambda_1 - \lambda_2)w y_1 y_2$. $\square$

Theorem 3 (Simplicity of eigenvalues). *Every eigenvalue is simple: its eigenspace is one-dimensional.*

Proof sketch. If y_1, y_2 are eigenfunctions for the same λ , their Wronskian $W = p(y_1 y_2' - y_2 y_1')$ is constant by (3) with $u = y_1$, $v = y_2$. The separated BCs force $W = 0$ at an endpoint, hence $W \equiv 0$ and y_1, y_2 are linearly dependent. $\square$

Remark 1 (Oscillation). *In the regular case, the n -th eigenfunction has exactly $n-1$ zeros in (a, b) (Sturm oscillation theorem). We do not prove this here, but it follows from Sturm comparison.*

4. Quick examples (reduction only)

Hermite on $[0, 1]$. $y'' - 2t y' + \lambda y = 0$, $y(0) = y(1) = 0$. Here $P = -2t$, so $p(t) = e^{\int -2t dt} = e^{-t^2}$ and

$$-(e^{-t^2} y')' = \lambda e^{-t^2} y \quad (q \equiv 0, w = e^{-t^2}).$$

Euler on $[1, e]$. $t^2 y'' + t y' + \lambda y = 0$, $y(1) = y(e) = 0$. Note $(ty')' = ty'' + y'$, so

$$-(ty')' = \lambda t^{-1} y \quad (p = t, q \equiv 0, w = t^{-1}).$$

Equivalently, with $x = \ln t$ and $u(x) = y(e^x)$ one gets $u'' + \lambda u = 0$ on $[0, 1]$.

Takeaway. Given $y'' + P y' + Q y = \lambda W y$ on a compact interval with separated endpoint conditions, multiplying by $p = e^{\int P}$ yields the self-adjoint Sturm-Liouville form (2). The Lagrange identity controls the boundary term, from which symmetry, reality, discreteness, orthogonality, and simplicity of the spectrum follow.